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Bubble dynamics in microchannel: An overview of the state-of-the-art

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Abstract The inception of the boiling, in a pool or flow boiling, is the formation of the vapor bubble at an active nucleation site that plays a crucial role in the boiling process and it becomes critical and unfolds many facets when channel size reduces to submicron. The detailed knowledge of the bubble dynamics is helpful in establishing the thermal and hydraulic flow behavior in the microchannel. In the current paper, bubble dynamics that include bubble nucleation at the nucleation site, its growth, departure, and motion along the flow in a microchannel(s) are discussed in detail. Different models developed for critical cavity radius favorable for bubble nucleation are compiled and observe that models exhibit large deviation. The bubble growth models are compiled and concluded that the development of a more generalize bubble growth model is necessary that would be capable of accounting for inertia controlled and thermal diffusion controlled regions. Bubbles at nucleation sites in a microchannel grow under the influence of various forces such as surface tension, inertia, shear, gravitational and evaporation momentum. Parametric analysis of these forces reckoned that the threshold between macro- to microchannel could be identify through critical analysis of such forces. Eventually, the possible impact of the various factors such as operating conditions, geometrical parameters, thermophysical properties of fluid on bubble dynamics in microchannel has been reported.

Keywords Microchannel · Flow boiling · Bubble growth · Elongated bubble · Confined bubble · Forces · Thermophysical properties

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Abbreviations

a_0:	Area (m^2)
A_{sf}:	Bubble surface area (m^2)
Bo:	Bond number
D:	Diameter (m)
D_h:	Hydraulic diameter (m)
F_b:	Buoyancy force (N)
f_i:	Interfacial friction factor (-)
F_i:	Inertia force (N)
F_m:	Evaporation momentum force (N)
F_s:	Surface tension force (N)
F_{sh}:	Shear force (N)
G:	Mass flux (kg/m^2s)
g:	Acceleration due to gravity (m/s^2)
h_{fg}:	Latent heat of evaporation (J/kg)
H:	Height of microchannel (m)

k: Thermal conductivity (W/mK)
L: Length (m)
m: Mass flow rate (kg/s)
N: Number of channels (−)
P: Pressure (N/m^2)
Pr: Prandtl number (−)
q'': Heat flux (W/m^2)
r, R: Radius (m)
t: Time (s)
T: Temperature (°C)
U_h: Homogeneous velocity (m/s)
U_G: Velocity of elongated bubble (m/s)
V_b: Volume of the bubble (m^3)
W: Width of microchannel (m)
y_b: Bubble height (m)

Greek

δ: Film thickness (m)
θ: Contact angle (deg)
ρ_l: Liquid density (kg/m^3)
ρ_v: Vapor density (kg/m^3)
σ: Surface tension (N/m)
α: Function of contact angle, void fraction (-)
β: Function of contact angle, bubble aspect ratio (-)
μ: Viscosity (Pa-s)

Subscript

a: Advancing
atm: Atmosphere
b: Bubble, bulk, buoyancy
c: Cavity
CHF: Critical heat flux
c/s: Cross section
in: Inlet
L, l: Liquid
lv: Liquid–vapor
max: Maximum
min: Minimum
r: Receding
sat: Saturation
sub: Subcooling
V, v: Vapor
w: Wall

1 Introduction

Rapid growth in usage of advanced information technologies (Internet, telecommunication services, simulation software etc.) has created unique yet critical demands on the power requirement to energize this network. Similarly, use of other applications like fusion reactor [1, 2], semiconductor laser [3] and advanced military avionics [4] have been increased rapidly due to their development and exploration of uses in diversified fields. Continuing market requirements for higher-speed access of aforementioned applications have resulted in dramatic increase in heat flux density. Heat generation rate in aforementioned applications has gone up to 10^2 W/cm^2 in densely packed integrated circuit (ICs) [5, 6] and laser mirror [7], 10^3 W/cm^2 in aviation and VLSI industry [4] and 10^4 W/cm^2 in fusion reactor and defense application [1, 2]. For an effective and efficient performance of the microprocessor, the temperature should be maintained below 85 °C [8, 9]. To obtain contradictory requirements of high heat dissipation and low temperature rise of the microprocessor or aforementioned applications, conventional cooling technologies such as natural or forced convection using air as working fluid are inadequate as this techniques are insufficient to provide heat removal capacity in excess of 1 W/cm^2 [10]. Air is widely used coolant in many low heat flux applications but its weak thermal transport properties greatly reduce its cooling potential compared to fluorochemical liquids and water under natural convection cooling strategies [11]. The forced convection cooling strategies show better heat transfer rate relative to natural convection, however, superior performance requires a phage change i.e. boiling of the liquid coolant. Up to 50% of an average air-cooled data center's energy consumption and carbon footprint is not caused by computing but by powering the necessary cooling systems to keep the processors from overheating [12]. Thus, use of the microchannel heat sink for cooling purposes of such sensitive devices is one of most promising solution [11, 13]. Single-phase liquid flow in microchannel heat sink is capable of removing heat flux over 500 W/cm^2 [14]. Thermal performance of the flow boiling in microchannels is impressively high due to the combined effect of very small hydraulic diameter and associated latent heat of evaporation. It can remove heat flux of more than 10^3 W/cm^2 [15]. Sekar et al. [16] compared the performance of the air cooled processor and microchannel cooled processor (both 3 GHz) and reckoned that former processor consumes around 19% less power compared to air cooled technique maintaining its temperature at 47 °C whereas air cooled

processor's temperature is 88 °C. Recently, IBM scientist demonstrated that liquid cooled microchannel heat sink is able to maintain super computer's (built for the Swiss Federal Institute of Technology (ETH) in Zurich as part of IBM's First-Of-A-Kind (FOAK) program) component temperature below 45 °C and claims that it consumes 40% less energy than air cooled system [17]. The pioneer study of Tuckerman and Pease [18] at Stanford University in the early eighties of the twentieth century addressed the applicability and superiority of the microchannel heat sink for electronic cooling. Thereafter, scientists across the globe are involved in research activities related with the purpose to understand and explore the thermo-hydraulic within microchannel heat sinks and deploy them in a variety of applications.

Enormous literature is available that covers the various aspects of the microchannel heat sink theoretically, experimentally as well as numerically. Following is the summary of the review articles presented by the different laboratories (and/or authors) emphasizing studies related to the microchannels heat sink. Papaustky and Ameel [19] reviewed a laminar single phase flow in microchannels. Kandlikar [20] reported fundamental issues related to flow patterns, heat transfer and pressure drop characteristics associated with microchannels. Kandlikar and Grande [21] discussed briefly about the basics of the microchannels and in details about micro-fabrication techniques. Hassan et al. [22] addressed single phase (liquid and gas) flow and flow boiling in microchannels. Steinke and Kandlikar [23] discussed possible implementation of various heat transfer enhancement techniques for microchannels, minichannels and microdevices. These techniques generally include flow transition, break up of boundary layer, entrance region modifications, vibration, electric fields, swirl flow, secondary flow and mixer. Thome [24] reviewed studies related to the two phase flow in microchannels. This consists of macro to microscale transition, two phase flow patterns, flow boiling heat transfer mechanisms and modelling based on thin film evaporation. Kandlikar [25] presented design considerations for the single phase liquid flow and flow boiling in microchannels and postulated that more efforts are needed to understand flow behavior in the microchannel. The flow patterns and flow patterns map under adiabatic and diabatic conditions are reviewed by Cheng et al. [26]. Their review contains a modeling of the different flow regime, heat transfer and pressure drop based on the flow regimes. Major limitation using boiling in cooling applications is critical heat flux (CHF). Roday and Jensen [27] and Das et al. [28] critically reviewed the CHF studies associated with microchannel. Lee et al. [29] studied the two phase pressure drop and heat transfer correlations in evaporative micro/minichannels. Agrawal and Singh [30] addressed the effect of heat flux and mass flux on the flow regimes, pressure drop, heat transfer, instabilities and CHF associated with a channel having a hydraulic diameter less than 1000 μm. Further they addressed an effect of the channel geometry and impact of dissolved air on the boiling. Morini et al. [31] discussed various techniques of the gas flow measurement in microchannel to predict accurate pressure drop and heat transfer by minimizing uncertainties associated with dimension and roughness measurement, pressure drop measurement, temperature measurement, local velocity, flow rate. Kandlikar et al. [32] provides detailed literature review of current research (up to year 2012) involvement addressing single-phase gas flow, enhancement in single-phase liquid flow and flow boiling, flow boiling instability, condensation, electronics cooling, and microscale heat exchangers. Kandlikar [33] addressed progress made in understanding the underlying mechanism associated with single phase liquid flow and flow boiling in microchannel. Talimi et al. [34] presented a comprehensive review on the numerical studies on hydrodynamics and heat transfer in non-boiling two-phase flows in small (mini and micro) scales. Baldassari and Marengo [35] presented fruitful information about non-dimensional numbers emphasizing more on Eotvos number, Weber number, Kandlikar numbers associated with microchannel. Mohammadi and Sharp [36] discussed various experimental techniques for bubble dynamics analysis in microchannels. Kadam and Kumar [37] provided in depth information of flow visualization, pressure drop, heat transfer and instabilities associated with microchannel heat sink. Bandara et al. [38] reviewed the non-boiling slug flow in microchannel and its intrinsic relationship with two phase heat transfer. Different instabilities associated with microchannel, effect of different parameters on flow instabilities and their mitigation techniques are recently addressed by Prajapati and Bhandari [39]. Abdollahi et al. [40] summarized thermal and hydrodynamics behavior under the influence of the film thickness, Peclet

number, working fluid and flow geometry of liquid–liquid two phase flow in microchannel. Sidik et al. [41] discussed an impact of passive techniques to improve the heat transfer performance of the microchannel heat sink. Karayiannis and Mahmoud [42] discuss the possible applicability of the flow boiling in microchannel heat sink for diversified fields, flow patterns and heat transfer mechanism, flow instability, effect of surface characteristics, effect of surface wettability, CHF.

Flow boiling in the microchannel has significant potential to capture and dissipate high heat flux due to the involvement of latent heat during the phase change process and high ratio of heat transfer surface area to volume of flowing fluid. However, deployment of the two phase microchannel seems challenging due to vapor bubbles and slugs induces instability leading to dry out and subsequent degradation in heat transfer rate [43–45]. Inception of the boiling in a microchannel (or in any boiling process) is the formation of the bubble at the nucleation site. In the open literature, no comprehensive review has been carried out that addresses exclusive bubble dynamics in the microchannel. As a matter of the fact, in the current paper, bubble dynamics in the microchannel are exclusively reviewed. Bubble dynamics in microchannel have been discussed in detail at different regimes such as bubble dynamics at nucleation sites, elongated and confined bubbles. Different models associated with bubble dynamics (before detachment and after detachment) presenting bubble growth, bubble departure, and bubble velocity in the microchannel are explored. The various forces acting on the bubble in the microchannel are identified and their quantitative impacts have carefully been discussed. Eventually, the possible impact of the various parameter such as operating condition (heat flux, mass flux, subcooling), microchannel geometry, roughness, thermophysical properties (in particular latent heat evaporation, surface tension, liquid and vapor density, dynamic viscosity) have been discussed in detail.

The conditions for the nucleation of the bubble are favorable liquid and surface temperature (i.e. wall superheat) and availability of the cavity. After the inception of the bubble, its growth undergoes through unconfined growth to confined growth if it is not detached from the nucleation site. Under the certain situations, the bubble detached from the nucleation site before experiencing the confinement effect are otherwise confined and then detached or lead to elongation at the nucleation site. The bubble growth and departure of the bubble are greatly influenced by the forces induced due to flow, the interaction between fluids and surface (surface tension effect), gravitational, and evaporation. Also, at all stages of the bubble dynamics, thermophysical properties play an important role. Detailed review considering the above fact has been carried out. Figure 1 shows the intrinsic relation between hydro-thermal characteristics (heat flux, pressure drop), and bubble dynamics. In the current paper, attention has been focused on the dotted box as shown in Fig. 1. The current review helpful to researchers of the field to build or develop bubble dynamics based mechanistic modelling of hydro-thermal characteristics associated with the microchannel.

2 Nucleation criteria

Boiling in the microchannel is expected to occur in either saturated liquid or subcooled liquid at the entry.

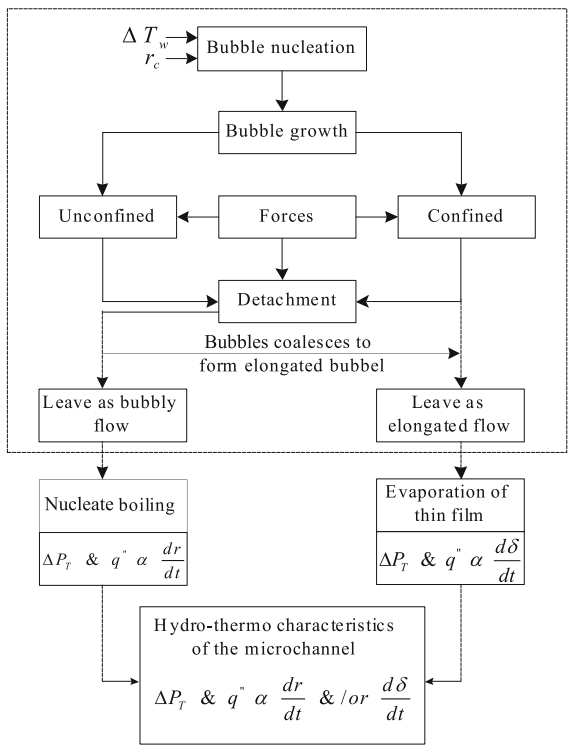

Fig. 1 Dependency of the bubble dynamics and hydro-thermal characteristics

In both conditions, bubble inception at the nucleation site is an important consideration to sustain the boiling in the microchannel in order to dissipate the large heat. The first time, Hsu [46] proposed the criteria for the bubble inception at the nucleation site for the pool boiling. According to his theory, for nucleation of the bubble two conditions have to be satisfied. First is that the temperature of the liquid, in vicinity of the heat transfer surface, must be equal and above the saturation temperature of the liquid corresponds to the surrounding pressure. Depending upon the type of manufacturing process adopted, a number of nucleation sites of different shapes and sizes are present at the heat transfer surface. However, under given operating conditions bubbles are not incepted at all these entire nucleation sites. Thus, the second condition is that the radius (r_c) of the nucleation site must fall within the range of $r_{c,min}$ and $r_{c,max}$. Such nucleation sites are called active nucleation sites. These two criteria are important to decide whether a nucleation site is active to incept and sustain the bubble. After satisfying these two conditions, a bubble nucleate at the nucleation site and appear at the cavity mouth. Although nucleation theory has been proposed for the pool boiling, it is expected that there is no considerable difference from the conventional theories for large and narrow diameter channels.

Consider that the bubble is nucleated and appears at the cavity mouth as shown in Fig. 2a. This bubble experiences the unbalance force due to the pressure difference between the inside (P_V) and outside (P_L) environment of the bubble which is balanced by the surface tension force given by Eq. (1).

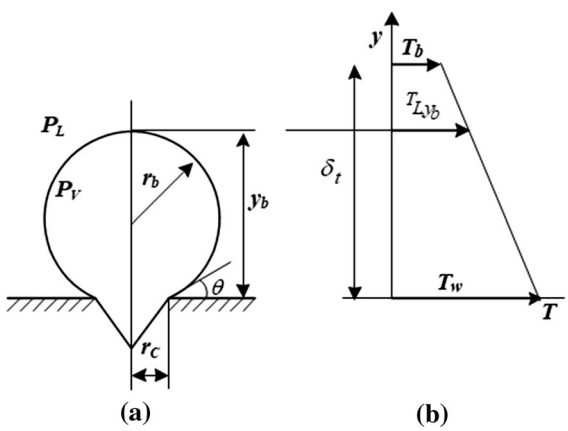

Fig. 2 Onset of nucleate boiling

$$p_v - p_L = \frac{2\sigma}{r} \tag{1}$$

At the vicinity of the nucleation cavity, a thermal boundary layer of thickness δ_t is formed with linear temperature variation away from the heat transfer surface as shown in Fig. 2b. Hence, the temperature of a liquid at the bubble tip (i.e. $y = y_b$) is;

$$T_{L,y_b} = T_w - \frac{y_b}{\delta_t}(T_w - T_b) \tag{2}$$

According to the Clausius-Clapyron equation, the pressure difference across the bubble is related to h_{fg}, v_v, v_L, and T_{sat} as follows;

$$p_v - p_L = \frac{h_{fg}\left(T_{Lsat(P_v)} - T_{sat}\right)}{T_{sat}(v_v - v_L)} \tag{3}$$

This equation is reduced to Eq. (4) as the specific volume of the vapor is much higher than the specific volume of the liquid.

$$p_v - p_L = \frac{h_{fg}\rho_v\left(T_{Lsat(P_v)} - T_{sat}\right)}{T_{sat}} \tag{4}$$

Equating Eq. (1) and Eq. (4), we get,

$$T_{Lsat(P_v)} = T_{sat} + \frac{2\sigma}{r}\frac{T_{sat}}{h_{fg}\rho_v} \tag{5}$$

To sustain the nucleation of the vapor bubble the minimum temperature requirement at any point on the liquid–vapor interface is $T_{L,yb} \geq T_{L,Sat(pV)}$. In other words, the bubble will sustain when its tip temperature (at y_b) is greater than the saturation temperature of the liquid corresponds to neighborhood vapor pressure. Combining Eqs. (2) and (5) yields the condition for nucleating cavities of specific radius:

$$\frac{2\sigma}{r}\frac{T_{sat}}{h_{fg}\rho_v} + \frac{y_b}{\delta_t}(T_w - T_b) + T_{sat} - T_w = 0 \tag{6}$$

From the bubble geometry, Fig. 2a, $r_b = r_c$ / sinθ and $y_b = r_c$ (1 + cosθ)/sinθ, substituting these into Eq. (6), upon rearranging we get,

$$\frac{r_c^2}{\delta_t}\frac{(1+\cos\theta)}{\sin\theta}(\Delta T_{sat} + \Delta T_{sub}) - \Delta T_{sat} r_c + \frac{2\sigma T_{sat}}{h_{fg}\rho_v}\sin\theta = 0 \tag{7}$$

where, ΔT_{sat} (= $T_w - T_{sat}$) and ΔT_{sub}(= $T_{sat} - T_b$) are the degree of wall superheat and degree of subcooling, respectively. Roots of the quadratic Eq. (7) are the minimum cavity radius ($r_{c,min}$) and maximum cavity radius ($r_{c,max}$) required for the bubble nucleation obtained by Davis and Anderson [47].

Various authors [46–50] have adopted similar phenomena with different relations between y_b and bubble radius r_b as summarized in Table 1.

The square root term in all Equations compiled in Table 1 is necessarily positive; else no nucleation site will be effective. Waiting time plays an important role during the process of the bubble inception. Satisfying the condition $r_{c,min} < r_c < r_{c,max}$ is not sufficient for the cavity to be an active nucleation site. For example, there are two cavities close to each other that satisfy the criteria ($r_{c,min} < r_c < r_{c,max}$), the cavity becomes active having a shorter waiting time [46] and the other one would be effective but non active. All other nucleation sites which fall beyond $r_c < r_{c,min}$ and $r_c > r_{c,max}$ are not effective.

Figure 3 shows the variation of the critical cavity radius (r_c) for the experimental data of Liu et al. [50]. It is observed that the cavity radius is sensitive to the wall temperature. With a slight variation in the wall temperature, the critical cavity radius significantly increases for all considered (Table 1) equations which also broaden the range of the active nucleation sites. In other words, a large number of cavities become activated and promote boiling in a vigorous manner inside the microchannel. It is also observed that previously derived equations for the critical cavity radius show a significant variation with experimental data. It is reckoned that the effect of the mass velocity of the liquid has not been considered while modelling the cavity radius which certainly has an impact on the active nucleation size in the microchannel.

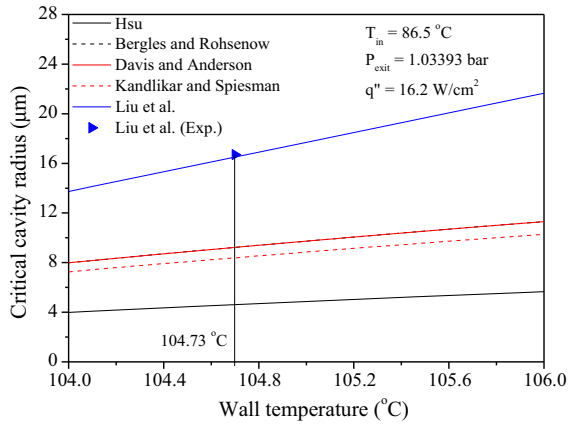

Fig. 3 Variation of the critical cavity radius with wall temperature

3 Bubble growth sequence

The very first step of the boiling starts with bubble formation at the nucleation site and boiling in a microchannel is no exception. Bubble dynamics have

Table 1 Correlations for Cavity radius

Author	Relation between y_b and r_b	Correlation
Hsu [46]	$y_b = 1.6\ r_b$	$\{r_{c,\min}, r_{c,\max}\} = \frac{\delta_t}{4}\left(\frac{\Delta T_{sat}}{\Delta T_{sat}+\Delta T_{sub}}\right) \times \left(1 \mp \sqrt{1 - \frac{12.8\sigma T_{sat}(\Delta T_{sat}+\Delta T_{sub})(1+cos\theta)}{\rho_v h_{fg}\delta_t \Delta T_{sat}^2}}\right)$
Bergles and Rohsenow [47]	$y_b = r_b$	$\{r_{c,\min}, r_{c,\max}\} = \frac{\delta_t}{2}\left(\frac{\Delta T_{sat}}{\Delta T_{sat}+\Delta T_{sub}}\right) \times \left(1 \mp \sqrt{1 - \frac{8\sigma T_{sat}(\Delta T_{sat}+\Delta T_{sub})}{\rho_v h_{fg}\delta_t \Delta T_{sat}^2}}\right)$
Davis and Anderson [48]	$y_b = r_b$ $(1 + \cos\theta)$	$\{r_{c,\min}, r_{c,\max}\} = \frac{\delta_t \sin\theta}{2(1+\cos\theta)}\left(\frac{\Delta T_{sat}}{\Delta T_{sat}+\Delta T_{sub}}\right) \times \left(1 \mp \sqrt{1 - \frac{8\sigma T_{sat}(\Delta T_{sat}+\Delta T_{sub})(1+\cos\theta)}{\rho_v h_{fg}\delta_t \Delta T_{sat}^2}}\right)$
Kandlikar and Spiesman [49]	$y_b = 1.1\ r_b$	$\{r_{c,\min}, r_{c,\max}\} = \frac{\delta_t \sin\theta}{2.2}\left(\frac{\Delta T_{sat}}{\Delta T_{sat}+\Delta T_{sub}}\right) \times \left(1 \mp \sqrt{1 - \frac{8.8\sigma T_{sat}(\Delta T_{sat}+\Delta T_{sub})(1+\cos\theta)}{\rho_v h_{fg}\delta_t \Delta T_{sat}^2}}\right)$
Liu et al. [50]		$\{r_{c,\min}, r_{c,\max}\} = \frac{\sin\theta}{(1+\cos\theta)}\left(\frac{T_w + \frac{2\sigma(1+\cos\theta)q''_w}{\rho_v h_{fg}\ k_l} - T_{sat}}{2\frac{q''_w}{k_l}}\right) \mp \left(\frac{\sqrt{\left(T_w + \frac{2\sigma(1+\cos\theta)q''_w}{\rho_v h_{fg}\ k_l} - T_{sat}\right)^2 - 4\frac{2\sigma(1+\cos\theta)q''_w}{\rho_v h_{fg}\ k_l}T_w}}{2\frac{q''_w}{k_l}}\right)\frac{\sin\theta}{(1+\cos\theta)}$

a great influence on the thermal and hydraulic behavior of the flow in a microchannel. the first time, Hsu [46] postulated that bubble formation takes place on heat transfer surface once surface temperature reaches at least equal to/greater than the saturation temperature of the liquid corresponding to the surrounding pressure. As the pressure in the liquid (vapor pressure) exceeds the surrounding pressure (atmospheric pressure), the growth of the vapor bubble from the surface is possible at a small nucleation cavity [51]. Bubble dynamics consist of the bubble nucleation, growth at nucleation site, departure from the nucleation site, and its motion along with the flow in the downstream direction. Complete bubble dynamics in the microchannel is presented schematically in Fig. 4.

Once a bubble is incepted at the nucleation site present on the heat transfer surface, it grows for a certain duration at its birthplace and departs from the nucleation site under the situation when various forces acting on the bubble are unbalanced [52]. The diameter at which the bubble depart from the nucleation site is called bubble departure diameter and plays important role in bubble dynamics in the microchannel. Under certain circumstances (in case of high heat flux or low mass flux or very small sized microchannel), the bubble grows to the microchannel diameter before detachment [53–55] experiences the confinement from the microchannel wall. The confinement process starts when bubble diameter reaches the smallest dimension of the microchannel called partial confinement. Then the vapor bubble starts expanding in another dimension of the microchannel. The state at which the vapor bubble occupies the entire cross section and the start of elongation in the axial flow direction is then bubble said to be fully confined [55, 56]. In case of circular and square microchannel, partial and complete confinement states are coincide. For the circular cross section of the microchannel, the bubble experiences both partial and complete confinement when its diameter reaches the diameter of the circular microchannel. The cross sectional dimensions i.e. width and height of square microchannel are the same unlike other cross sections (for example-rectangular). When bubble diameter reaches equal to any dimension indicating that bubble also occupies another dimension of the microchannel. This implies that partial confinement of the bubble coincides with complete confinement in the case square cross section. However, in certain cases (low heat flux/high mass flux/ large microchannel), the bubble may depart from the nucleation site before it experiences partial confinement and moves along the flow direction [50]. Situ et al. [57] observed that bubble after departure slides along the heat transfer surface for a certain period and then detached into bulk flow in macrochannel. Similarly, Owhaib et al. [58], Fu et al. [59] revealed that bubble slides along the wall of the minichannel before liftoff in the bulk flow after departure from the nucleation site. However, in the case of the microchannel, it is difficult to identify the bubble sliding because of its very small size. While

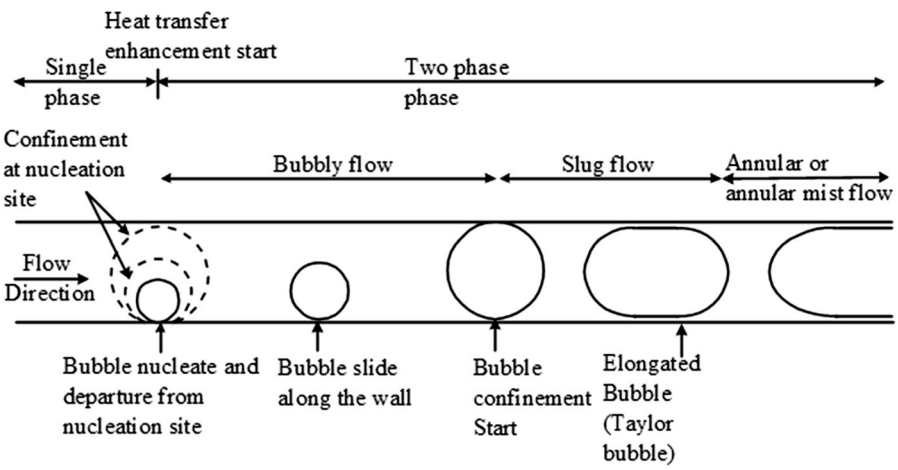

Fig. 4 Complete bubble dynamics in microchannel [37]

sliding along the wall or moving with the flow, its size grow rapidly due to continuous heat addition from the wall. This growing bubble tries to occupy the microchannel's cross section experiencing partial or complete confinement. After confinement, the vapor bubble starts expanding into the flow direction leading to elongated bubbly (Taylor bubble) flow or annular flow when downstream end moves out of the microchannel. The elongated bubble finally leaves the microchannel or may lead to flow reversal phenomena [60, 61] at high heat flux or low mass flux value (Fig. 5).

Table 2 summarizes the bubble dynamics studies associated with the microchannel heat sink. From Table 2, it is observed that the bubble dynamics in the microchannel have been studied mostly for the water as a working fluid. However, it is equally important to study the bubble dynamics including nucleation followed by different regimes for different refrigerants as working fluid. The thermophysical properties of the refrigerants are quite different from the water (in particular latent heat of evaporation and surface tension). Thus, bubble dynamics of the refrigerants would help in-depth understanding of the hydro-thermal behavior inside microchannel heat sink. Further, it is observed that the bubble often encounters confinement at the nucleation site itself before detachment from the nucleation site due to small channel size. Thus, it can be concluded that the evaporation of thin liquid film spread over a considerable length of the microchannel is the dominating heat transfer mechanism. However, activation of a large number of nucleation sites over the length of the microchannel [53, 85] nucleate boiling mechanism could dominate heat transfer. Thus, the dominance of governing heat transfer mechanisms in microchannels is still unclear and laid a topic of intense research. In-depth understanding of the bubble dynamics in microchannel would help in giving further insight into the dominating heat transfer mechanism. However, vigorous boiling (under high heat flux or low mass flux) in a microchannel made it impossible to identify the dominating heat transfer mechanism.

4 Bubble growth regime at nucleation site

The vapor bubble at the nucleation site grows under the influence of two different growth controlled mechanisms, namely inertia controlled and thermal

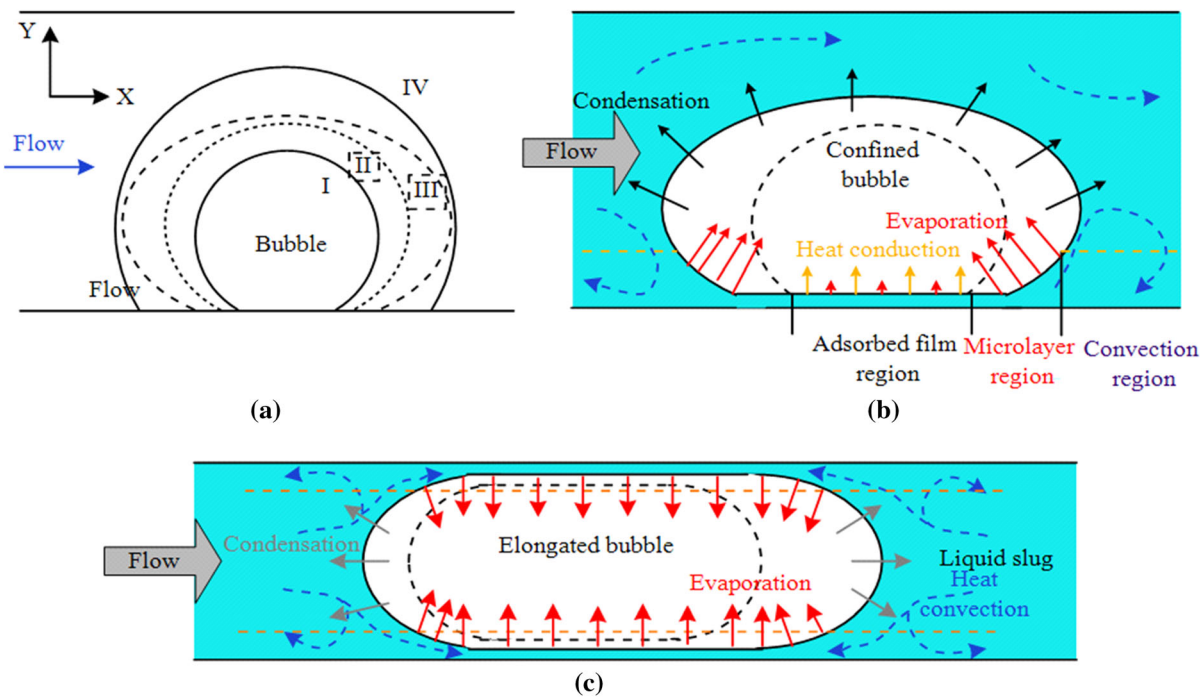

Fig. 5 **a** bubble deformation; **b** heat transfer mechanism for confined bubble; **c** heat transfer mechanism of elongated bubble [79]

Table 2 Summary of the microchannel geometry, operating conditions, and concluding remark(s) from experimental observations

Author	Year	Microchannel geometry (W, H, D) (μm), N (-), L (mm)	Operating conditions G (kg/m^{2}s), q" = kW/m^2	Confinement observed	Remark
Jiang et al. [62]	2001	Triangular, silicone D_h = 26 N = 34, 35, L = 98.6	Water, Q_l = 0.0315 – 0.063 ml/min, q = 5.1 – 29.9 W	No	The microchannel size may affect the bubble formation and growth mechanism. Bubbles may be generated due to dissolved gases since the device temperature is less than the saturation temperature.
Steinke and Kandlikar [63]	2003	Rectangular, copper W = 214, H = 200 N = 6, L = 57.15	Water, G = 115 – 467 q" = 41.7 – 632	Yes	The smallest bubbles (D = 15 to 60 μm) and the largest bubbles (D = 125 to 200 μm) move with a low velocity of 0.184 m/s as their behavior has been influenced by the microchannel wall. Whereas, the velocity of the largest bubble hindered by an increase in friction and drag of the microchannel wall. Medium size bubble (D = 60 to 200 μm) moves much faster (1.28 m/s) as they move to the middle of the flow experiencing less resistance from the microchannel wall.
Lee et al. [52]	2003	Rectangular, silicone W = 120, H = 14	Water, Q_l = 0.0315 – 0.063 ml/min, q = 0.17 – 2.04 W	Yes	The bubble grow in all directions at an active nucleation site until filling the microchannel cross section. After occupying the cross section, it starts stretching downward direction forming a long stem however still attached to the nucleation site and departs under unbalanced force condition. The liquid film is present in between the bubble and microchannel corners.
Lee et al. [64]	2003	Rectangular, silicone W = 100 H = 50 N = 1	Water, G = 33.14 ΔT_{wall} = 5.6^{0}C	Yes	Linear bubble growth at the nucleation site.
Lee et al. [53]	2004	Trapezoidal, silicone D_h = 41.3, N = 1, L = 28	Water, G = 170 – 477 q" = 1.47 – 449	Yes	Bubble growth is classified as isotropic (spherical in shape) and anisotropic (distorted shape due to microchannel confinement and flow drag). The local flow field alters the growth cycle under the same operating condition at the nucleation cavity under consideration. Bubble growth is linear suggesting that bubble growth is controlled by inertia. However, in some cases growth is explosive.

Table 2 continued

Author	Year	Microchannel geometry (W, H, D) (μm), N (-), L (mm)	Operating conditions G (kg/m^{2}s), q" = kW/m^2	Confinement observed	Remark
Li et al. [54]	2004	Trapezoidal, silicone D_h = 47.7, N = 2, L = 28	Water, G = 105 – 555 q" = 12 – 202	Yes	Bubble growth is linear suggesting that bubble growth is controlled by inertia and follows t^2 indicating growth is diffusion controlled. Reverse flow exists frequently in almost all cases of two parallel microchannels compared to the single microchannel [53]. In slug flow, larger pressure in the liquid slug pushes the bubble tail to be flatten compared to the front end that experiences less resistance of liquid since both, interface and liquid, are moving in the same direction.
Liu et al. [50]	2005	Rectangular, copper W = 275, H = 636 N = 25, L = 25.4	Water, G = 498 – 899 q" = 100.7 – 731.6	Yes	Bubble growth at the nucleation site is linear and departs from the nucleation site without experiencing confinement. The contact angle has a significant role at bubble nucleation and it decreases exponentially from inception until departure.
Balasubramanian and Kandlikar [65]	2005	Rectangular, copper W = 990, H = 207, N = 6, L = 63.5	Water, G = 112 – 120, q" 208 – 316	Yes	Bubble nucleated at an upper wall fills the width of the microchannel forming liquid film (approximately 50 μm thick) on the microchannel's sidewall. After confinement, the bubble grows rapidly due to increased evaporation. Bubble nucleation in the thin liquid film.
Huh and Kim [66]	2006	Rectangular, polydimethylsiloxane W = 100, 200 H = 107, 100 N = 1, L = 40	Water, G = 77.5 – 309.8 q" = 180 – 500	Yes	Elongated bubble shows continuous growth due to the evaporation of liquid film trapped between the vapor core and heated surface. An elongated bubble is distorted, asymmetric and experiences the sliding movement along with the flow due to the effect of the flow field while bubble still anchored to the heated surface.
Lee and Son [67]	2008	Rectangular, W = H	Water, P = 1 atm	Yes	The thin liquid layer exists between the bubble and microchannel corner. The thermal boundary layer has distorted in the presence of the bubble at the nucleation site and observed a very high heat flux region in the vicinity of the bubble wall contact line.

Table 2 continued

Author	Year	Microchannel geometry (W, H, D) (μm), N (-), L (mm)	Operating conditions G (kg/m^{2}s), q" = kW/m^2	Confinement observed	Remark
					Bubble slides along the wall before detachment.
Zhaun and Wang [68]	2010	Rectangular, copper W = 275, H = 636 N = 25, L = 25.4	Water, G = 498 – 899 q" = 100.7 – 731.6	Yes	At low incipient heat flux, surface tension controls the bubble growth and heat transfer at the bubble interface has been dominated by the Marangoni flow induced due to temperature gradient. At higher incipient heat flux, a temperature gradient is low and the Marangoni effect turns weak, thus growth is controlled by high incipient heat flux.
Barber et al. [55]	2010	Rectangular, glass W = 4000 H = 400 N = 1, L = 80	FC-72, m = 1.33 × 10–5 kg/s q" = 4.26	Yes	The rapid growth of the bubble leads to sharp pressure fluctuation in inlet and outlet plenum. Pressure fluctuation is periodic since bubble nucleation to elongation of the slug growth recurs in a cyclic manner. The bubble grows freely and rapidly in spherical and symmetrical shape while moving downstream until experiencing the confinement from the smallest dimension and then deform into sideways filling larger cross section. After complete confinement, the tail of the vapor slug recedes towards the inlet.
Krishnamurthy and Peles [69]	2010	Rectangular, silicone W = 200, H = 243 N = 5, L = 10 Circular fins D = 100, pitch = 400 N_f = 120	HFE-7000 G = 350 – 827 q" = 100 – 1100	No	Bubble departure frequency decreases with an increase in subcooling and departure diameter. Bubble nucleate at the rear wake side of the first fin (larger diameter) in the boiling region which alters the flow field hydrodynamics in downstream and causes bubbles to nucleate (smaller diameter) at different locations (frontal stagnation point) on the downstream fins. The rear side bubble grew asymmetrically due to enhanced recirculation of flow, at some point it necked close to the fin surface and finally detached. However, the frontal bubble slides along the fin circumference while growing and lifts off from the rear side of the fin.

Table 2 continued

Author	Year	Microchannel geometry (W, H, D) (μm), N (-), L (mm)	Operating conditions G (kg/m^{2}s), q" = kW/m^2	Confinement observed	Remark
Lee et al. [70]	2011	Rectangular, silicone W = 50, 100 H = 46, 48, 100 N = 1, L = 64	Water, G = 7.97 – 33.3 ΔT_w = 2 – 6^{0}C	Yes	The bubble grows at a natural nucleation site, reaches to microchannel size, departs, sweeps along the microchannel and interacts with adjacent nucleated bubbles (causes agglomeration and merging with the subsequent bubbles) in the downstream direction. The bubble grows (proportional to t^3) to occupy microchannel and expands promoting annular flow at artificial nucleation site. The bubble grows explosively with a slight increase in wall superheat trapping thin liquid film, in both cases.
Wang et al. [71]	2011	Rectangular, borosilicate W = 4000 H = 400 N = 1, L = 80	FC-72 G = 8.75 – 175 q" = 0 – 15.03	Yes	Bubble first occupies the smallest dimension, and then the largest dimension followed by the length of microchannels (up- and downstream) leading to the flow reversal and rewetting (temporary dry out or pre-mature CHF [72]). Bubble nose growth rate has high acceleration compared to tail because of different temperature and surface tension gradients exist at both contact lines (at the tail and at the nose).
Edel and Mukherjee [73]	2011	Rectangular, brass W = 201 H = 266 N = 1, L = 24.7	Water, G = 122, 144 Ts = 100.9 – 130.7	Yes	Bubble growth rate after confinement is higher than before confinement due to an increase in the surface area of the liquid–vapor interface. Rapidly growing confined bubble suppress the growth of the nearby bubble due to local pressure spike. The onset of the bubble elongation would shift towards upstream due to increased resident time of the higher temperature liquid.
Gedupudi et al. [74]	2011	Rectangular, copper W = 380 H = 1500, 1600 N = 1, L = 40	Water, G = 747 q" = 173, 210	Yes	Irregularities in bubble shape after confinement due to the presence of small bubbles in the liquid film and attached to the window by surface tension. Tail velocity is nearly equal to incoming liquid velocity in absence of inlet compressibility. On the other hand, inlet compressibility allows meager shifting of the bubble from the nucleation site towards upstream subsequently grows in both upstream and downstream direction.

Table 2 continued

Author	Year	Microchannel geometry (W, H, D) (μm), N (-), L (mm)	Operating conditions G (kg/m^{2}s), q" = kW/m^2	Confinement observed	Remark
Bogojevic et al. [75]	2013	Rectangular, silicone W = 150, H = 273 N = 40, L = 15	Water, G = 71 – 204 q" = 138 – 537	Yes	Stage I, inertia controlled; stage II- diffusion controlled; stage III- rapid growth as interface enters into the highly superheated liquid region of adjacent walls. However, the lifetime of the bubble at sidewall is longer than the bubble at the bottom wall due to low superheat near the sidewall. The explosive growth of a confined bubble promotes boiling instabilities.
Yin et al. [76]	2014	Rectangular, copper W = 500 H = 1000 N = 1, L = 10	Water, G = 33.6 – 66.7 q" = 28.9 – 65	Yes	The growth rate of the confined bubble decreases with an increase in bubble size due to the increase in growth resistance when the bubble interface approaches close to the microchannel wall. The rapid growth of the confined bubble at the nucleation site compared to the unconfined bubble and attributed it to the evaporation of the thin liquid film.
Yin et al. [77]	2015	Rectangular, copper W = 1000 H = 1000 N = 1	Water, G = 86.7 – 130 q" = 84.9 – 129.8	No	In the slow sliding stage, constant bubble size indicates the rate of evaporation at interface in the superheated liquid near to the microchannel wall balanced by the rate of condensation at bubble interface, which is in contact with subcooled bulk liquid. Bubble size reduces (condensation effect override evaporation effect) in accelerated sliding stage due to possible migration of the bubble into the main subcooled flow.
Yin et al. [78]	2016	Rectangular, copper W = 500 H = 1000 N = 1, L = 10	Water, G = 33.3 – 66.7 q" = 28.9 – 69.2	Yes	Vapor bubble top surface (at nucleation site) deforms from spherical cap to flat interface before it reaches up to the wall due to additional force exerted by the microchannel wall. The onset of the microchannel wall confining effect starts when the first occurrence of the slope decrement of maximum local void fraction and a sudden dramatic reduction in bubble aspect ratio occurred.
Yin et al. [79]	2016	Rectangular, copper W = 500, H = 1000 N = 1, L = 10	Water, G = 20, 40 q" = 64.7 – 104.2	Yes	Figure 5(a) shows bubble growth at nucleation site; stage I- truncated shape, stage II- the influence of the wall confinement, stage III- expansion axial direction forming

Table 2 continued

Author	Year	Microchannel geometry (W, H, D) (μm), N (-), L (mm)	Operating conditions G (kg/m^{2}s), q" = kW/m^2	Confinement observed	Remark
					microlayer, stage IV- retaining to truncated shape due to rapid evaporation of microlayer [80, 81]. Heat transfer mechanism at the time of bubble confinement ((1) heat conduction through underneath thin film, (2) intense evaporation of the increased microlayer and (3) convection around bubble) and elongation (extensive evaporation of the microlayer [82–84]) as shown in Fig. 5(b) and (c), respectively.
Prajapati et al. [85]	2017	Rectangular, copper 1) Uniform W = 400, H = 750 N = 12, L = 25.7 2) Diverging W_i = 300 W_o = 520 H = 750 N = 12, L = 25.7 3) Segmented Dh = 522	Water, G = 100 – 350 q" = 10—350	Yes	The bubble growth trend is the same but the growth rate is different in all three configurations under the same operating condition. Bubble growth has been suppressed in diverging microchannel compared to uniform microchannel as it takes a longer time to occupy the same diameter. After confinement in segmented channels the bubble grows into a zigzag manner suppresses flow reversal and also wall temperature oscillation and pressure fluctuation is low.
Kingston et al. [56]	2018	Circular, borosilicate D = 500 N = 1, L = 42	HFE-7100 G = 200 – 800	Yes	The smooth nature of glass (generally microchannel(s) are smooth due microfabrication process irrespective of material [52]) and high wetting characteristics of HFE-7100 delayed the inception. Any change in the local pressure field near to the bubble can be experienced throughout the microchannel leading to large pressure and temperature spike in up- and down- [86, 87] and promotes rapid bubble growth instability.

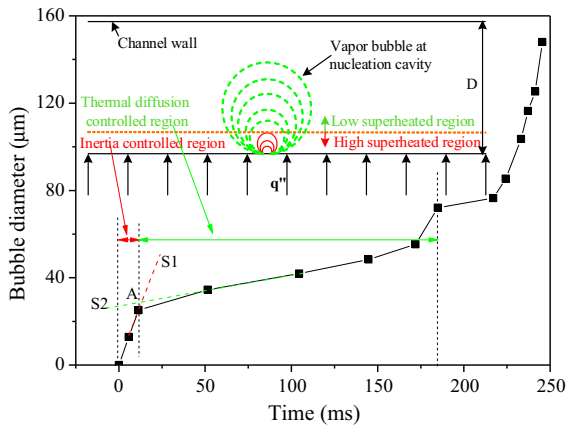

Fig. 6 Bubble growth regimes at nucleation site [75]

diffusion controlled mechanisms [54, 75, 79]. At the very early stage of bubble growth, its size is very small and close to the heat transfer surface. Also, it is surrounded by a liquid having a higher degree of superheat. Thus, the rate of the evaporation is higher, and the bubble growth is governed by the reaction force of the surrounding liquid at the bubble interface. This stage is called inertia controlled region. After reaching bubble diameter to a certain size, where the bubble interface is in contact with a low degree of superheated liquid, the reaction force (inertia force) turns weak and the latent heat of evaporation is supplied by the thermal diffusion (between vapor–liquid interface and surrounding liquid). As the vapor bubble is surrounded by a higher degree of liquid superheat, its growth rate is higher in inertia controlled region compared to the thermal diffusion controlled region.

Figure 6 schematically presents the bubble growth at the nucleation site in the microchannel considering inertia controlled and thermal diffusion controlled mechanisms. In inertia controlled region, the bubble grows very rapidly up to A (indicated by slope S1) and bubble growth slows down in thermal diffusion controlled region after A (indicated by slope S2). The intersection of the line S1 and line S2 (i.e. point A) indicates transitional boundary of the bubble growth controlled regions. Bogogevic et al. [75] observed that around 50% growth of the bubble took place in the inertia controlled region in a short time (around 25 ms) and the remaining growth of the bubble took around 175 ms. Beyond this time, the bubble interface enters into a highly superheated liquid region near the adjacent walls. This results in rapid evaporation and hence rapid bubble growth. The similar bubble growth trend was observed in inertia and thermal diffusion controlled regions by Yin and Jia [79]. Meaning, a large amount of the heat is being transferred from the heat transfer surface in inertia controlled region compared to thermal diffusion controlled region [88]. If the bubble can be departed before (before point A) it enters into thermal diffusion controlled region, bubble frequency would increase promoting a higher heat transfer rate.

5 Bubble dynamics modeling

Bubble growth inside the microchannel gives an insight into the heat transfer and pressure drop characteristics. The modelling of the bubble dynamics is categorized as bubble growth at the nucleation site, departure diameter models, modelling of the elongated bubble.

5.1 Modelling of nucleated bubble

The bubble growth rate at the nucleation site or while moving along with flow after detachment is an indicator of the rate of evaporative heat carried away by the fluid. Hence, it is of vital importance to predict bubble growth inside the microchannel. Rayleigh equation, as given by Eq. (8), can be utilized for obtaining a homogeneous growth rate in the superheated bulk medium [51, 89, 90] with the assumption that the bubble is surrounded by superheated liquid.

$$\frac{dr}{dt} = \left[\frac{2}{3}\frac{\Delta P_v}{\rho_l}\right]^{1/2} = \left[\frac{2}{3}\frac{(T_w - T_{sat})h_{fg}}{T_{sat}v_{lv}\rho_l}\right]^{1/2} \tag{8}$$

However, in the case of microchannel, it gives a very large bubble growth rate [53, 54, 91]. Under the certain operating conditions, predicted bubble growth rate by the Rayleigh equation is 4.89 m/s. However, the experimental bubble growth rate under similar operating conditions for the microchannel is 2.37×10^{-4} m/s [53]. In the case of the microchannel, the bubble experiences a wall constrained at least in a transverse direction where the temperature gradient is expected to be high. This results in rapid evaporation only in a region close to the wall, and thus bubble growth is much smaller in microchannel than

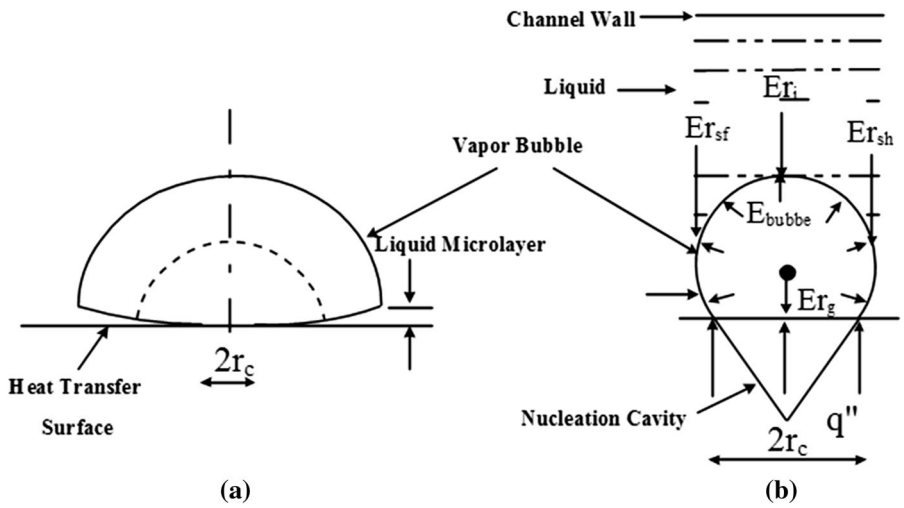

Fig.7 Nucleated bubble **a** with microlayer [92], **b** under energy distribution [94]

predicted by the Rayleigh equation. Cooper [82] proposed a model to predict the growth rate of the bubble nucleated at a heated surface. The fundamental assumption is that the evaporation of the liquid microlayer formed underneath the bubble being responsible for the bubble growth as shown in Fig. 7a. According to his model, bubble radius is a function of the Jacob number, Prandtl number, and square root of time as given by Eq. (9).

$$r = 2.5\frac{Ja}{\mathrm{Pr}_l^{1/2}}(\alpha_1 t)^{1/2} \tag{9}$$

where, Jacob number (Ja) is $\left(=\frac{\rho_l C_{pl}(T_w - T_{sat})}{\rho_v h_{fg}}\right)$ and Prandtl number (Pr) is $\left(=\frac{\nu_L}{\alpha_v}\right)$. However, from the experimental study [53], it confirmed that the possibility of formation of the liquid microlayer may not be

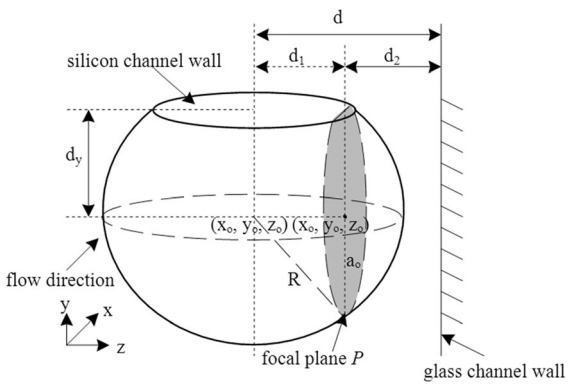

Fig. 8 Bubble geometry reconstruction [93]

there as bubble growth shows a linear function of time and not the square root.

Wang et al. [93] proposed a hybrid method to predict bubble growth rate in the microchannel. The information of experimental bubble images and the velocity flow field around the bubble measured using micro particle image velocimetry (μPIV) are used in conjunction with three-dimensional simulation to reconstruct the bubble geometry. The focal plane P, at a distance of d_2 from the glass channel wall (as shown in Fig. 8) which gives an area with radius a_0, has been adjusted in such a way that the predicted three-dimensional velocity vectors match with velocity vectors generated by the μPIV. Ones, these two velocity vectors match, then bubble geometry can be estimated having a radius, $R\left(=\sqrt{d_1^2 + a_0^2}\right)$.

This approach can be useful for the reconstruction of three-dimensional bubble geometry, location and related three-dimensional velocity field during bubble inception and growth. However, the applicability of the model for flow boiling is ambiguous since the model is validated for the bubble nucleated due to the outgassing of the liquid.

Kadam et al. [94] developed an energy based bubble growth model for predicting the bubble growth rate at the nucleation site in a microchannel as presented by Eq. (10). They have critically analyzed various resistive effects experienced by the bubble. They assumed that the energy supplied to the vapor phase is utilized for bubble growth and overcoming

the various effects such as surface tension, flow inertia, shear, gravitational, and evaporation momentum effect as shown in Fig. 7b.

$$q''\frac{A_c}{A_{ch}}\frac{\gamma}{2}t = \left[g(\rho_l-\rho_v)\frac{\pi\cdot\alpha\cdot\varphi}{2}\right]\cdot\left(r^2-r_{\min}^2\right) + \left[\sigma\cdot(2\cdot\pi\cdot\beta)+\frac{\mu\cdot G}{4\cdot\rho_l}\pi\cdot\beta\right](\ln(r)-\ln(r_{\min})) + \left[\pi\cdot\alpha\cdot\rho_v h_{fg}+\frac{G^2}{\rho_l}\frac{\gamma}{2}+\left(\frac{q''}{h_{fg}}\right)^2\frac{\pi\cdot\beta}{\rho_v}\right]\cdot(r-r_{\min}) \quad (10)$$

The bubble growth at the nucleation site is divided between the inertia controlled region and thermal diffusion region. However, it is identified that available models fail to predict distinctly different bubble growth rates in these two regions at the nucleation site. Hence, there is a need to develop a more generalized bubble growth model which can account for these regions. This can be possible by accounting for the effect of the thermal boundary layers formed near the heat transfer surface and possible condensation from the top surface of the bubble. In the early stage, the thermal boundary layer in close vicinity of the surface may be responsible for extensive bubble growth. The possible condensation from the top surface of the bubble may play an important role in a later bubble growth region, which suppresses the bubble growth in the subcooled region away from the surface. Based on energy based bubble growth model, Kadam and Kumar [88, 95] have identified non-dimensional energy ratio, E_1, to demarcate inertia and thermal diffusion controlled regions as given by Eq. (11) and observed that the transition value of E_1 varies from 400 to 2400 (this range is based on experimental data of Ref. [75, 79]). Hence, they concluded that rigorous experimentation is required to set its transition criteria (for water and other fluid as a working fluid).

$$E_1 = E_{bubble}/Er_{sf} = \frac{\rho_v h_{fg}}{2\sigma}r\frac{\alpha}{\beta} \quad (11)$$

From the rigorous analysis, it is identified that models proposed for the prediction of bubble growth at nucleation site in the microchannel are validated for water as working fluid, and hence there is clear ambiguity in the applicability of these models for fluids such as refrigerants (such as R134a, R236fa, R245fa, R123, R410a) and other fluid (FC and HFE series). The flow boiling in microchannel(s) have been extensively studied using a refrigerant(s) [96–101] and other fluid [102–105]. Thermophysical properties of the refrigerant(s) are varying significantly compared to the water, which may alter the thermo-hydro behavior of the flow inside the microchannel. Hence, it is important and recommended to carry out rigorous experimentation that emphasizing the bubble dynamics at the nucleation site using refrigerants (or FC and HFE series fluids) which also facilitate data for model's validations.

5.2 Modelling of bubble departure diameter

Very few authors have studied and proposed bubble departure diameter correlation for the microchannel. Qu and Mudawar [106] proposed a model for the bubble departure diameter in microchannel based on force balance. They considered that drag force and surface tension force are balanced at the time of bubble departure [107].

$$R_d = \frac{\sigma}{u_c^2}\frac{C_s}{C_d}\frac{1}{\rho_f}f(\theta) \quad (12)$$

where, u_c is the liquid velocity at half-way point from the nucleation site to the bubble tip, C_d (= $24/Re_b$ for $4 < Re_b < 20$; 1.22 for $20 < Re_b < 400$) is drag coefficient, C_s (= $58/(\theta_0 + 5) + 0.14$) is empirical coefficient, $f(\theta)$ is equilibrium contact angle. For the bubble nucleated at the side and bottom wall, $f(\theta) = \frac{\pi\sin\theta_0(\cos\theta_r-\sin\theta_a)}{\pi-\theta_0+\cos\theta_0\sin\theta_0}$. For the bubble nucleated at the corner, $f(\theta)$ is given by Eq. (13) for $0 \le \theta_0 \le \pi/4$; $f(\theta)$ is given by Eq. (14) for $\pi/4 \le \theta_0 \le \pi/2$.

$$f(\theta) = \frac{2\pi\sin\theta_0(\cos\theta_r-\sin\theta_a)}{\pi-2\theta_0+2\cos\theta_0\sin\theta_0} \quad (13)$$

$$f(\theta) = \frac{2\left[\pi-\arccos\left(\frac{\cos\theta_0}{\sin\theta_0}\right)\right]\sin\theta_0(\cos\theta_r-\sin\theta_a)}{\frac{3\pi}{4}-\theta_0+\cos\theta_0(\cos\theta_0+\sin\theta_0)} \quad (14)$$

where, θ_0, θ_a (= $\theta_0 + 10^\circ$) and θ_r (= $\theta_0 - 10^\circ$) are equilibrium, advancing and receding contact angle, respectively. Li et al. [54] proposed an empirical correlation for the prediction of the bubble departure diameter following Levy's approach [108] as given by Eq. (15).

$$R_d = C_1 \left(\frac{\sigma L}{\Delta P}\right)^{1/2} + C_2 \tag{15}$$

where, C_1 (= 7.63 X 10^4) and C_2 (= 22.6) are empirical constant and obtained from experimental data. They observed that 85% of data of both Ref. [53, 54] falls within the error band of ± 30%.

Few efforts have been made to study the bubble departure in the microchannel, theoretically as well as experimentally. Hence, generalized applicability (under a wide range of operating conditions and different fluids) of the proposed departure diameter models are ambiguous.

5.3 Modelling of elongate bubble

Some authors have modelled a confined and elongated bubble to understand the thermo-hydro behavior inside the microchannel. Jacobi and Thome [109] modelled a pair of elongated bubble and liquid slug (two zone model) after the bubble swept away from

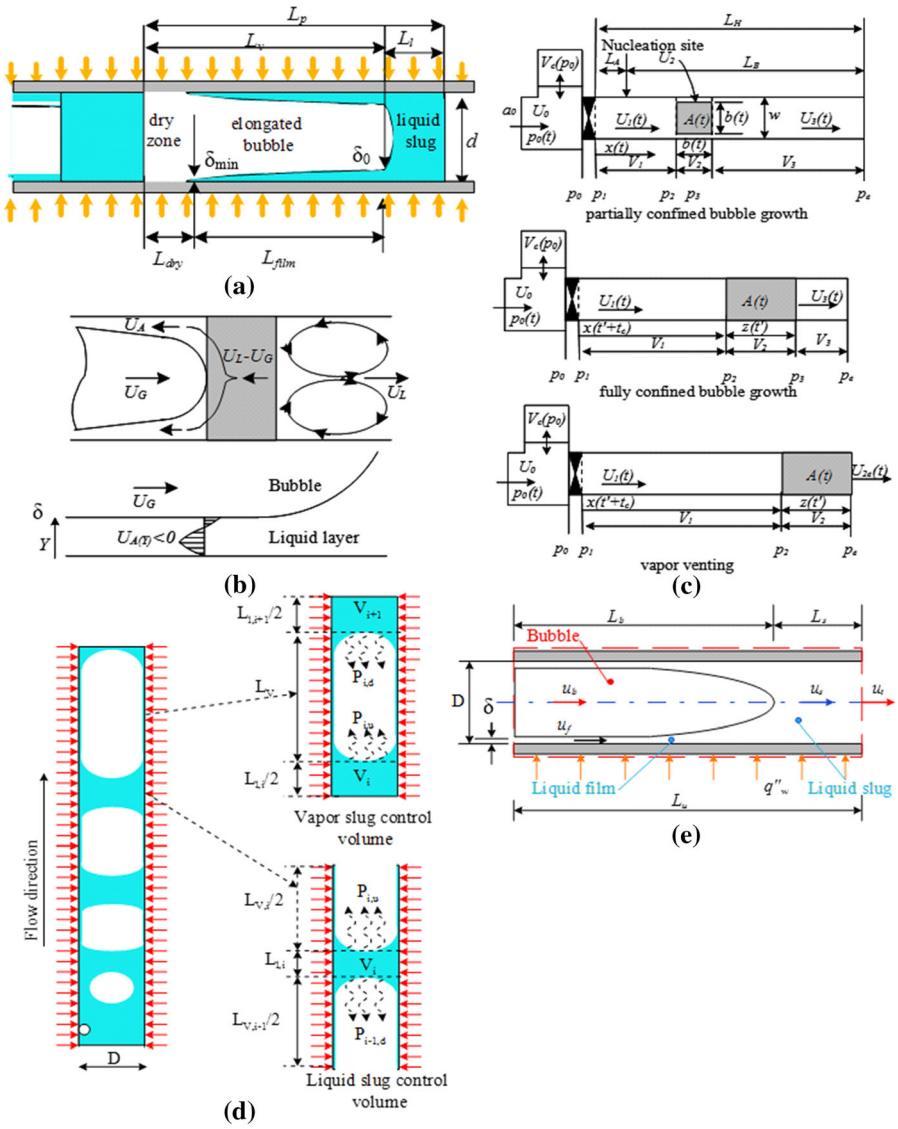

Fig. 9 **a** Three zone heat transfer model [110]; **b** Typical velocity profile in liquid layer trap between an elongated bubble and microchannel wall [113]; **c** Stages of bubble growth [74]; **d** Control volume of liquid and vapor slug [115]; **e** Fully developed slug flow unit for flow boing in a horizontal microchannel [118]

the nucleation site to understand heat transfer characteristics in this region. The thin liquid film has been formed when liquid slug passes. Due to evaporation, its thickness decreases and continues until the next liquid slug passes. Thus, they assume that the evaporation of the thin liquid film trapped between the elongated bubble and the microchannel wall is dominating the heat transfer mechanism. Thome and co-author [110, 111] developed and validated three zone model (liquid slug, elongated bubble, dry vapor if film dries out as shown in Fig. 9a) to understand the local heat transfer behavior in microchannel during slug flow regime. They assume that at a fixed location the process is executed as liquid slug passes, elongated bubble passes and vapor slug passes if thin liquid film dried out, and cycle repeats itself till the next liquid slug passes. Liquid layer thickness reduces due to evaporation adding vapor to the elongated bubble and thus it further elongates contributing to a sudden dramatic increase in heat transfer coefficient. Initial film thickness at the bubble's nose has been obtain using Eq. (16) following experimental data of Moriyama and Inoue [112]. The homogeneous two phase flow approach has opted to obtain void fraction and two phase velocity.

$$\delta_0 = d \times C_{\delta_0}\left(3\sqrt{\frac{\nu_l}{U_p d}}\right)^{0.84}\left[\left(0.07Bo^{0.41}\right)^{-8}+(0.1)^{-8}\right]^{-1/8} \tag{16}$$

where, C_{δ_0} empirical correction factor obtained from experimental data.

Agostini et al. [113] have developed the model to determine the elongated bubble velocity, which depends on its length, using mass, momentum and energy conservation equations. They schematically explained that the velocity profile in the liquid layer confined between the elongated bubble and microchannel wall is necessarily negative as shown in Fig. 9b. This is attributed to a faster velocity of the bubble compared to liquid slug, which causes some liquid to be evacuated by the liquid layer. However, such a hypothesis has not been validated experimentally in the microchannel. They obtained an expression for the elongated bubble velocity (U_G) bubble in microchannel in terms of its length (L_G) as given by Eq. (17).

$$U_G = \frac{d \times \gamma}{1+\frac{C}{Co}}\frac{1-\exp\left(\frac{-2f_i L_G}{d}\right)}{2f_i} + U_h \tag{17}$$

where, $C = -0.58$, $\gamma = \frac{4q''}{d \times h_{fg} \times \rho_G}$, Co (= 0.58), f_i, U_h is confinement number, interfacial friction factor and homogeneous velocity respectively. They observed that the proposed correlation of bubble velocity shows good agreement with experimental data and depicts 90% data within error band of ± 20%. Further, out of their modeling, a parametric impact study is provided on the bubble relative velocity ($U_G - U_h$). They observed that for given heat flux, bubble velocity increases with an increase in mass flux and microchannel diameter and a decrease in saturation temperature. The collision/merging of the bubbles results in a transition from elongated bubble flow to an annular flow towards the microchannel's outlet. Revellin et al. [114] proposed a model for the bubble collisions in a microchannel by extending the model of Agostini et al. [113] with the fundamental assumption that longer trailing vapor bubble collides with a smaller leading vapor bubble due to its higher velocity. This assumption gives a condition as given by Eq. (18).

$$L_{L,i+1} - \frac{\Delta z}{U_{G,i+1}}\left(U_{G,i} - U_{G,i+1}\right) \leq 0 \tag{18}$$

where, subscripts L and G are for the liquid and vapor phase and i and $i + 1$ correspond to the trailing bubble and leading bubble respectively. $L_{L,i+1}$ is the length of a liquid slug between two elongated bubbles.

Gedupudi et al. [74] developed 1-D model for the confined bubble growth to check the effect of the inlet compressibility. The analysis is divided into three stages namely: partial confinement growth, fully confined growth and vapor venting as shown in Fig. 9c. They observed that pressure difference between inlet and outlet plenum significantly increase during fully confined growth compared to partially confined growth.

Li and Hrnjak [115] modelled bubble dynamics in a single microchannel. Bubble growth is divided into partial confinement and complete confinement. In partial confinement, the period between bubble inception until departure has been governed by a well-known correlation proposed by Plesset and Zwick [51] and given by Eq. (19).

$$r(t) = Ja\sqrt{\frac{12\alpha_l t}{\pi}} \tag{19}$$

A similar approach during partial confinement has been opted by Thome and co-author [109, 110, 113]. In the full confinement stage, the control volume is a combination of one liquid slug and one vapor slug as shown in Fig. 9d. Due to the evaporation, the length of the liquid slug decreases leading to further elongation of the vapor slug. It is assumed that when the length of the liquid slug is less than 5% of the hydraulic diameter of the microchannel, two vapor slugs coalesce. Mass, momentum and energy equations have applied to both liquid and vapor slug to obtain the relation between bubble velocity and heat flux as given by Eq. (20).

$$V_{i+1} - V_i = \frac{4q''\left(L_{v,i} + \frac{L_{l,i}}{2} + \frac{L_{l,i+1}}{2}\right)\left(1 - \frac{\rho_v}{\rho_l}\right)}{\rho_v D h_{fg}} \tag{20}$$

The model has been validated with experimental data of Ref. [116, 117] and shows good agreement.

Recently, Younes et al. [118] developed a model to predict the slug length, liquid film velocity and liquid slug velocity in fully developed slug flow region (Fig. 9e) considering the contribution of the applied heat flux and surface tension using mass, momentum and energy transport equations. The momentum equation is expressed as shown by Eq. (21),

$$\begin{aligned}&\left[\rho_g\alpha_g\left(\frac{\rho_g\alpha_g}{\rho_f} - 1\right) + 2\rho_f f_f\left(\frac{L_u}{D}\right)\right](u_b - u_t)^2\\&-\left(\frac{\rho_g\alpha_g}{\rho_f}\right)\left(\frac{4q''_w}{h_{fg}}\right)\left(\frac{L_u}{D}\right)(u_b - u_t)\\&-\left(\frac{1}{\rho_f}\right)\left(\frac{\alpha_g}{1-\alpha_g}\right)\left(\frac{4q''_w}{h_{fg}}\right)^2\left(\frac{L_u}{D}\right)^2\\&+4\tau_w\left(\frac{L_u}{D}\right) + 4\left(\frac{\sigma}{D}\right)\left(\frac{1}{\sqrt{\alpha_g}} - 1\right) = 0\end{aligned} \tag{21}$$

The reduction of the liquid slug length is given by Eq. (22) and an increase in the length of the elongated bubble is given by Eq. (23).

$$\begin{aligned}\frac{dL_s}{dt} = &\left(\frac{\rho_g\alpha}{\rho_f}\right)(u_b - u_t) + (1-\alpha)(u_f - u_t) + (u_s - u_t)\\&-\left[\left(\frac{\rho_g}{\rho_f} - 1\right)\alpha + 1\right]\frac{dL_b}{dt} - L_b\left(\frac{\rho_g}{\rho_f} - 1\right)\frac{d\alpha}{dt}\end{aligned} \tag{22}$$

$$\frac{dL_b}{dt} = 4A_c\left(\frac{R_s}{R_b}\right)^2\left[(t^+ + 1)^{1/2} - (t^+)^{1/2}\right] \tag{23}$$

where, α is cross sectional void fraction, t^+ is dimensionless time parameter, $\frac{d\alpha}{dt} = \left(\frac{4q''_w}{\rho_f h_{fg} D}\right)\alpha^{1/2}$, $t^+ = \left(\frac{A_c}{B_c}\right)^2 t$, $A_c = \left[\frac{b\rho_g h_{fg}\Delta T}{\rho_f T_{sat}}\right]^{1/2}$ and $B_c = \left[\frac{12Ja^2 D_f}{\pi}\right]^{1/2}$, D_f is liquid thermal diffusivity. They observed that bubble frequency (u_b/L_u) increases with vapor quality and attains peak at vapor quality approximately equal to 0.037 where liquid slug vanishes and the first stage of bubble coalescence begins. Further, they observed that the liquid slug length decreases in the isolated bubble zone and elongated bubble length increases by increasing vapor quality. This can be attributed to the thermodynamic and hydrodynamic effects that influence the bubble and liquid slug zones due to the liquid film evaporation that causes the elongation of the bubble, and the inertial forces that affect the bubble nucleation and liquid slugs in this zone.

6 Forces acting on the bubble and Bubble departure condition

For the simplified and mechanistic modeling of the heat transfer and pressure drop correspond to the flow boiling in the microchannel, the understanding of the bubble dynamics (nucleation, growth, departure, etc.) is important. The bubble incepted at the nucleation site experience various forces until departure. The bubble grows at the nucleation site to a certain diameter under the balance of various forces: forces that holds a bubble and forces that tries to detach the bubble. After reaching a certain diameter, the bubble is departed from the nucleation site under the unbalancing between holding and detaching forces. Various forces are considered during the boiling such as lift force, surface tension force, expansion due to the pressure difference, flow drag and temperature-induced drag, the force due to unsteady growth, evaporation

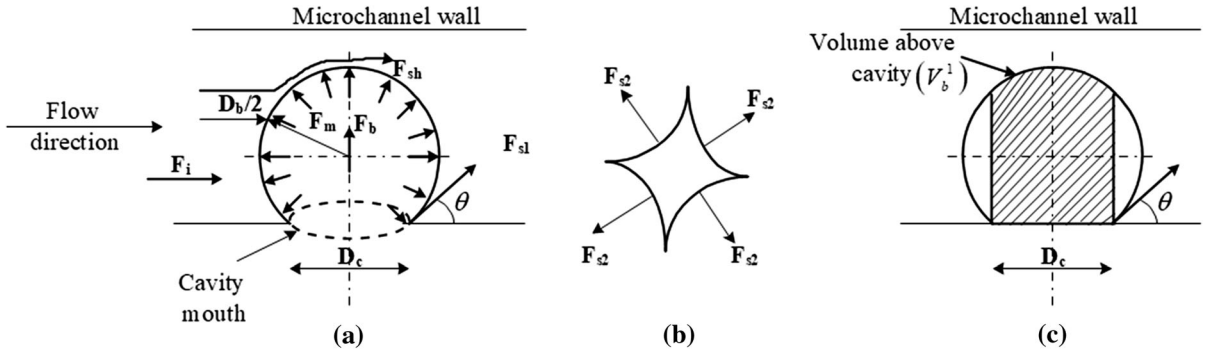

Fig.10 Forces acting on the vapor bubble in microchannel

momentum force, inertia force, shear force [119–122]. However, in the case of the microchannel, the dominant forces are inertia, surface tension, shear, buoyancy, drag and evaporation momentum force [94, 106, 123–127]. The existence of these forces on the bubble is presented in Fig. 10a.

The bubble is attached to the heat transfer surface at the nucleation cavity and experiences the surface tension force (contact pressure force) F_{s1} [123, 124, 128, 129]. The magnitude of F_{s1} can be considered as constant if the contact line (cavity diameter, D_c, not change) is not moving. But in reality, due to continuous heat addition, the contact line kept moving outward due to rapid evaporation at the liquid contact line. Further, bubble surface always possess curvature and thus surface energy. Consequently, the interface experience the force, F_{s2}, to maintain the curvature [94] as shown in Fig. 10b.

$$F_{s1} = \sigma \cdot \cos\theta \cdot \pi D_c \tag{24}$$

$$F_{s2} = \frac{G^*}{A_{sf}} \cdot D_b = \sigma \cdot D_b \tag{25}$$

where, D_c and D_b are the cavity diameter and diameter of curvature (bubble radius), G^* is Gibb's surface energy. The cavity diameter can also be expressed in terms of bubble diameter as $D_c = \sin\theta \cdot D_b$. Accordingly, the magnitude of the F_{s1} changes with change in the bubble diameter. The large magnitude of the F_{s1} is desirable since it increases the CHF limit by pulling liquid from the wet region towards the dry region [123]. Nevertheless, it may hold a bubble at the nucleation site for a longer time and this situation promotes a hot spot at the nucleation site as the heat transfer surface is in contact with vapor.

The liquid inside the microchannel flows with a certain velocity, which exerts a force, F_i, normal to cross section of the vapor bubble, and lose its momentum.

$$F_i = \frac{G^2}{\rho_l} A_{c/s} \tag{26}$$

A larger magnitude of the inertia force is desirable as it helps in detaching the bubble prior to experiencing the microchannel wall confinement, which is prone to flow reversal. Hence, it helps to increase the CHF limit and to avoid malfunctioning (instabilities) of the flow inside the microchannel. Similarly, moving fluid induces shear effect at bubble surface and hence experiences the shear force, F_{sh} [94, 123, 124, 128, 129]. The strong existence of the shear effect causes early departure of the bubble from the nucleation site.

$$F_{sh} = \frac{\mu G A_{sf}}{\rho_l D_b} \tag{27}$$

The density difference between the liquid phase and the vapor phase exerts the force called buoyancy force. This force can be expressed distinctively under two circumstances; F_{b1}, when the vapor bubble is completely immersed into liquid (entire volume of the bubble is considered) and F_{b2}, when the bubble is attached to the nucleation site where the volume of the non-shaded portion of the bubble (as shown in Fig. 10c) is considered [88, 130, 131].

$$F_{b1} = g(\rho_l - \rho_l)V_b \tag{28}$$

$$F_{b2} = g(\rho_l - \rho_l)V_b^1 \tag{29}$$

where, V_b is the volume of the vapor bubble and V_b^1 is the volume of the portion above the nucleation cavity. The resultant buoyancy force, F_b, will be given by Eq. (30).

$$F_b = g(\rho_l - \rho_v)\left(V_b - V_b^1\right) \tag{30}$$

The buoyancy force helps to detach the vapor bubble from the nucleation site. Hence, a large density difference between the liquid phase and the vapor phase is desirable.

As the liquid evaporates, the vapor leaves the evaporating interface rapidly and moves towards the center of the vapor bubble. The jetting of the vapor that leaves the interface exerts a force at the evaporating interface due to the change in momentum caused by the increase in velocity as the liquid phase changes to the vapor phase. This force resulted from the momentum change is called the evaporating momentum force (F_m), which is given by the product of evaporation mass flow rate and the vapor velocity leaving the interface, whose direction is opposite to that of vapor leaving into the bubble [125–129].

$$F_m = \left(\frac{q''}{h_{fg}}\right)^2 \frac{A_{sf}}{\rho_v} \tag{31}$$

Few authors studied the bubble departure from the nucleation site in microchannel [53, 54, 69, 87]. The

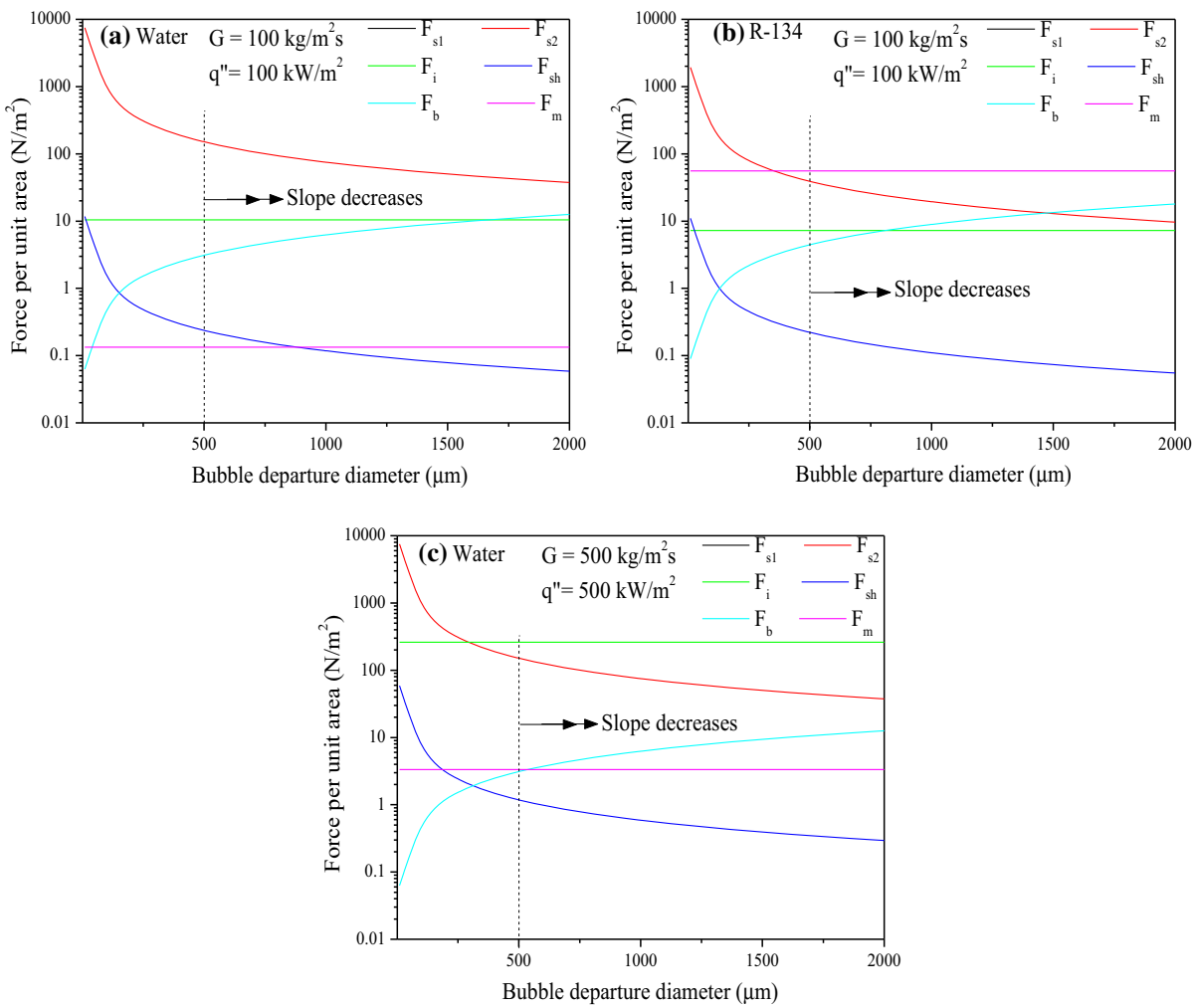

Fig. 11 Variation of the forces with bubble departure diameter

departure of the bubble from the nucleation site takes place when the summation of forces that are in favor of holding the bubble at the nucleation site turn weaker than the summation of forces those who try to pull out the bubble from nucleation site. Bubble departure diameter decrease with an increase in the mass flux and heat flux, and consequently bubble frequency increases. An increase in mass flux resulted in an increase in the magnitude of the associated forces acting on the bubble. Thus, the bubble is depart from the nucleation site at a small diameter. Similarly, with an increase in the heat flux, surface tension reduces due to temperature increase. In consequence, the surface tension force turns weak causing bubble to depart at a small diameter. The departure diameter can be estimated by balancing the forces that hold the bubble against the nucleation site and that tries to pull away the bubble.

Figure 11 shows the relative variation of the various forces acting on the bubble at atmospheric pressure, mass flux (100 and 500 kg/m^{2}s), and heat flux (100 and 500 kW/m^{2}s) for water and R134a as a working fluid. It is assumed that the bubble is on the verge of departure (on the verge of experiencing the channel wall's confinement hence bubble diameter is approximated as the hydraulic diameter of microchannel) and hence contact angle is zero. The forces are presented per unit area of the bubble cross section, which is equivalent to the microchannel hydraulic diameter. The shear effect and evaporation momentum effect seem less significant during the range of the bubble departure diameter for water as shown in Fig. 11a. This is in line with previous studies [123, 124]. However, near to CHF condition, the heat flux is high enough and thus evaporation momentum effect plays an important role at CHF condition [126, 128, 129]. Further, comparing the water and R134a (Fig. 11a, b) reduction in latent heat of evaporation facilitates the boiling at the interface. The buoyancy effect is very less at lower bubble diameter but turn out to be stronger after reaching the bubble to a certain departure diameter and overrides evaporation momentum, shear, and inertia effects. This implies that buoyancy effect should not be neglected while analyzing the flow boiling in the microchannel. The surface tension effect seems always dominating at departure. The inertia effect strongly depends on the mass flow rate of the liquid flowing through the microchannel and shows a significant impact at departure under considered operating conditions. Comparing Fig. 11a, c the evaporation momentum effect increases rapidly with an increase in heat flux. Similarly, inertia and shear effect increases with an increase in the mass flux. But, the high value of the mass flux prohibits rewetting during flow reversal and promotes instabilities. However, it is very difficult to set a specific trend of forces acting on the bubble since they are very sensitive to the operating conditions and thermophysical properties of the fluid.

In general, departure diameter is close to the hydraulic diameter of the microchannel and its shape no longer holds truncated shape (confinement effect). Thus, it is quite reasonable to consider departure diameter to demarcate minichannel and microchannel [68]. Interestingly, it is observed that the slope decrement or increment of the forces per unit area above 500 μm seems less significant under considered operating conditions and fluids. Further detailed analysis of these forces can be used to propose a threshold between macro to microscale.

Yin et al. [127] critically analyzed an evaporation momentum force and shear force acting on the meniscus of the elongated vapor bubble in the microchannel. Due to the higher evaporation rate near the heat transfer surface, the magnitude of the evaporation momentum force is highest near the vicinity of the contact line region. It is found that around 95% of heat transfer is taking place from 30% of the meniscus region near the contact line [132, 133] as shown in Fig. 12a and is termed as an effective evaporation region. The non-effective evaporation region is regarded as the meniscus of the elongated bubble where no evaporation happens. The effective evaporation momentum force ($F_{m,e}$) acting on the meniscus per unit length is given by Eq. (32).

$$F_{m,e} = 0.15\left(\frac{q''_w}{h_{fg}}\right)^2 \frac{(D-2\delta)}{\rho_v} \tag{32}$$

It is well accepted that liquid film exists between the elongated bubble and microchannel wall, evaporation of that film is the main source for further bubble elongation. This film replenishes due to the squeezing of the anterior liquid slug nearby the meniscus of the elongated bubble moving with high velocity. The liquid extracted due to squeezing is twice useful: liquid supply for effective evaporation region, and to

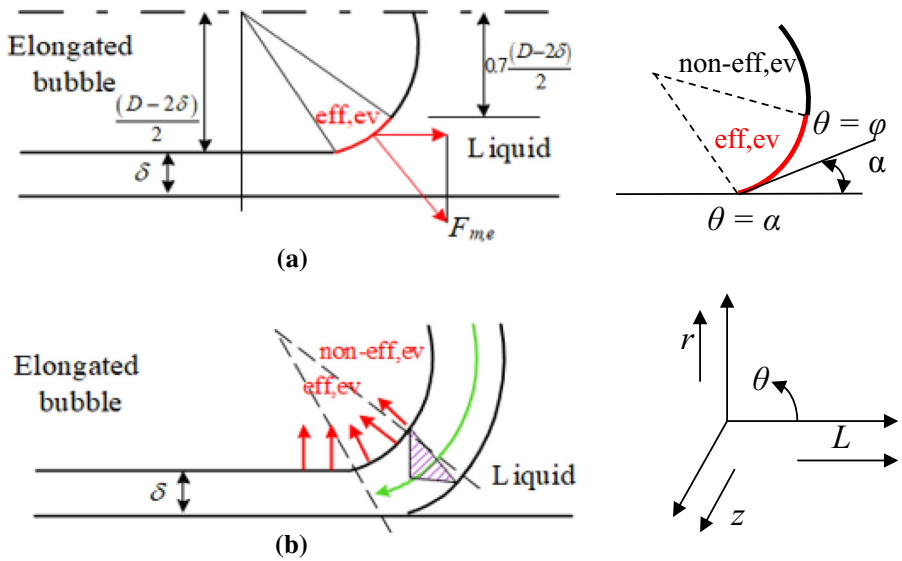

Fig. 12 Evaporation momentum and shear force acting on meniscus of the elongated [127]

sustain the liquid film. Due to the relative motion of liquid supplied for effective evaporation and elongated bubble, the meniscus of elongated bubble experience the shear force at an effective evaporation region and non-effective evaporation region (front meniscus of the bubble as shown in Fig. 12b). They only considered the axial component of the shear force due to the symmetrical characteristic of the meniscus of the elongated bubble. These two axial components of the shear force are given by Eq. (33) and (34), respectively.

$$F_{sh,eff-ev} = \frac{\left(\begin{array}{r}\mu q_w''(D-2\delta)^2(\varphi\sin\varphi+\cos\varphi-\cos\alpha-\alpha\sin\varphi)\\ -\mu q_w''(D-2\delta)L\cos\alpha(\sin\varphi-\sin\alpha)\end{array}\right)}{\rho_l\delta^2 h_{fg}\cos^2\alpha} \tag{33}$$

$$F_{sh,non-eff-ev} = \frac{2\mu(D-2\delta)\left(\frac{q_w''(D-2\delta)}{2h_{fg}\cos\alpha}(\varphi-\alpha)+\frac{q_w''L}{2h_{fg}}\right)[1-\sin\varphi]}{\rho_l\delta^2\cos\alpha} \tag{34}$$

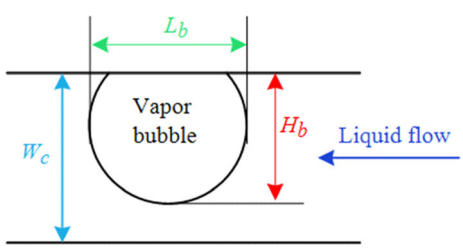

Fig. 13 Schematic of the bubble size parameters [78]

where, α is a contact angle, θ is a latitude angle ($\theta = \alpha \rightarrow$ start of the evaporating region; $\theta = \varphi \rightarrow$ end of the evaporating region and start of non-evaporating region) and $\varphi = \arccos(0.7\cos\alpha)$. Thus, the total shear force experienced by the evaporating and moving meniscus is a summation of the Eq. (33) and Eq. (34).

Yin and co-authors [78, 134] explained the qualitative significance of the wall confining force acting on a confined bubble which has a significant effect at microscale compared to macroscale. The significance is explained using maximum local void fraction ($\alpha = H_b/W_c$, represent bubble growth in bubble height direction) and bubble aspect ratio ($\beta = H_b/L_b$, indicate bubble deformation) as shown in Fig. 13.

They observed that the bubble aspect ratio decreases after a certain time (16 ms) due to the wall that introduces the confining effect at the bubble top interface causing its deformation from spherical cap to flat interface. Two forces are dominating the wall confinement: bubble growth force and liquid drag force. After certain growth of the bubble at the nucleation site, comparable to channel size, the resistance to push away the liquid near bubble top interface increases. This significantly increases the bubble growth force. Further, the high value of the maximum local void fraction represents that the bubble has occupied a considerable portion of the microchannel. This leads to a dramatic rise in liquid drag force along with bubble height. The increase in

the magnitude of both of these effects jointly creates wall confining force which slow down the movement of bubble top interface towards the microchannel wall. Bubble experience wall confining force right from its inception but its impact slowly increases when the bubble top interface moves toward the microchannel wall. Thus, it is important to study its quantitative influence during bubble growth in a microchannel.

7 Effect of the operating parameters on bubble dynamics

It is of vital importance to investigate the effect of the various parameters on bubble dynamics to understand boiling in the microchannel. Bubble nucleation and its growth are influenced by several parameters such as surface roughness, heat flux, mass flux, thermophysical properties of the liquid, liquid and surface interaction, channel geometry.

7.1 Heat flux

Heat flux has a great influence on bubble dynamics, as it is one of the decisive parameter to ensure the inception of the bubble at the active nucleation site. The wall temperature increases with an increase in heat flux and the bubble inception take place when $T_w \geq T_{sat}$. Further, an increase in wall heat flux may result in the activation of a large number of small nucleation sites [135] and also the inception of the boiling (onset of nucleate boiling) may tends to shift towards the inlet of the microchannel [85, 136, 137]. An increase in the bubble nucleation density results in an increase in the nucleate boiling heat transfer rate [135, 136]. At low heat flux range, the bubble grows linearly at the nucleation site, whereas, the bubble grows exponentially under high heat flux conditions due to the enhanced rate of evaporation [53, 54, 70]. Moreover, bubble growth may suppress at high heat flux conditions since ONB shifted towards the inlet of the microchannel. The movement of the ONB close to the inlet leads to an increase of the local saturation temperature and corresponding local pressure, which consequently suppresses the bubble growth at the nucleation site in the microchannel heat sink. Secondly, this situation more likely to occur in the subcooled boiling case in which inlet temperature is below saturation temperature, and thus cooling of the bubble top surface takes place to suppress the bubble growth.

7.2 Mass flux

An increase in the flow rate of the liquid may delay the boiling process and perhaps ONB shifted away from the inlet toward downstream. As well accepted, the rate of heat transfer in single phase liquid flow in the microchannel increases with an increase in mass flux. Thus, it does not allow surface temperature to be raised equal to or above the saturation temperature of the fluid in the upstream region. As flow progresses, wall temperature increases to enough degree to initiate the boiling. Thus, it can be concluded that with an increase in mass flux, ONB shifts to downstream direction. Furthermore, an increase in mass flux results in bubble growth reduction because; (1) reduction in the rate of evaporation [46], (2) increase in local convective heat transfer from bubble surface to surrounding liquid [79]. An increase in the mass flux also plays a great role in bubble departure from the nucleation site and enhances the bubble frequency [75]. Forces that are proportional to mass flux, namely inertia and drag, try to detach bubbles early from the nucleation sites due to their increased magnitude as the mass flux increases.

7.3 Inlet liquid subcooling

Flow boiling in microchannel seems less influenced by the degree of inlet subcooling [56]. However, an increase in the degree of subcooling may shift the bubble inception to a downward direction due to higher temperature difference between incoming bulk fluid and wall temperature, which gives rise to a single phase heat transfer rate. Yin et al. [75] recognized that the occurrence of the bubble confinement is postponed with an increase in the degree of subcooling due to enhanced condensation at the bubble interface. The high degree of subcooling reduces the growth rate of the confined bubble. This is attributed to the overriding condensation effect at the meniscus compared to the evaporation of the thin liquid film trapped between the vapor bubble and the microchannel wall. In the case of the state of flow reversal resulted from rapid bubble growth, a higher degree of subcooling reduces the rate of evaporation at the liquid–vapor interface and favors to minimize the upstream velocity of the

interface. This may result in a reduction in the intensity of the pressure spike generated in the upstream direction and thus reduces the duration of instability associated with rapid bubble growth.

7.4 Microchannel geometry

The geometry of the microchannel also plays a great role in bubble dynamics. Prajapati et al. [85] observed that in segmented microchannel, bubble elongation follows a zigzag path flowing from one channel to other that result in enhanced mixing and heat transfer rate compared to the uniform and diverging microchannel. Besides, the bubble tries to envelop fins but the sharp front edge of the rhomboid fin tries to break the elongated bubble and helps to avoid the formation of a long slug. This behavior allows the bubble to move out of the microchannel easily by facilitating more area for better heat transfer. Corner of a non-circular cross section of the microchannel(s) acts as a host for the active nucleation site [138] and enhances the heat transfer rate at low vapor quality. However, the situation may become reverse at higher vapor quality (annular flow) since surface tension force attenuates liquid film more effectively. This result in early local dry out in non-circular microchannel compared to microtube.

7.5 Roughness

Bubble formation starts at the active nucleation site whose shape and size are a function of the surface roughness of the heat transfer surface. This implies that roughness has a great influence on bubble dynamics during the boiling process, pool or flow boiling. A certain roughness is always desirable to activate the boiling process under certain favorable conditions. When surface roughness increases beyond a certain limit, it exerts a barrier to the contact line movement and prohibits the spreading of the bubble on the heat transfer surface. This may lead to the generation of the hot spot at the nucleation site. The surface roughness decides the ability of the surface to be remains wetted with liquid. Fouling of the surface may block the active nucleation site, which reduces the heat transfer rate by prohibiting the formation of the bubble due to agglomeration of the scale in cavities.

7.6 Thermophysical properties

Thermophysical properties of the working fluid have a significant impact on the flow boiling process [37, 88] and it is clear from Fig. 10. Generally, bubble dynamics have been extensively studied for the water as a working fluid and few addresses other cooling fluids such as FC-72, HFE-7000, HFE-7100. However, limited studies are available pertinent to the bubble dynamics using refrigerant as a working fluid in microchannel, though they are extensively studied. Table 3 present the thermophysical properties of some of these fluids. The important thermophysical properties associated with the boiling such as latent heat evaporation, liquid and vapor density, liquid viscosity and surface tension varies significantly from one working fluid to other and expectedly have a considerable impact on bubble dynamics in the microchannel.

Low latent heat of evaporation activates the nucleation site in abundance even at low heat flux. Whereas, boiling process become unstable when heat flux increases due to spiraling evaporation of the liquid leading to the rapid bubble growth at the nucleation site. This may trigger the blockage of the microchannel and thus promote flow reversal. Further, this situation can prone to the early occurrence of the CHF (as in the case of the refrigerant, CHF value is lower compare to the water [139]). Further, evaporation of

Table 3 Comparison of thermophysical properties of different fluids at 1 bar pressure

Fluid	ρ_l (kg/m^3)	ρ_v (kg/m^3)	H_{fg} (kJ/kg)	σ (N/m)	μ (10^{-4} Pa-s)
Water	958.6	0.590	2257.4	0.0590	2.83
FC-72	1680.0	13.240	88.0	0.0105	6.40
HFE-7100	1399.0	9.870	111.6	0.0103	3.61
R134a	1377.5	5.193	217.2	0.0152	3.80
R600a	594.2	2.792	365.4	0.0142	2.29
R290	581.2	2.387	425.9	0.0157	1.98

the liquid film, trapped between the elongated vapor bubble and microchannel wall, reduces with an increase in enthalpy of vaporization and hamper the heat transfer performance. Contrary, this would reduce the possibilities of the complete dry out of the liquid film.

The large value of the surface tension may delay the bubble nucleation due to the existence of the strong adhesive force between the heat transfer surface (i.e. cavity mouth) and fluid, and that holds the bubble at the active nucleation site (upon nucleation) for longer duration. This situation may lead to the possibility of bubble confinement at the nucleation site (generally observed in the case of water). Such situation can further lead to the generation of the hot spot at the nucleation side since large degree of superheat is required to initiate the bubble inception. On the other hand, a large temperature gradient exists at a higher degree of superheat, which reduces surface tension (Marangoni effect or thermo-capillary convection [140, 141]). Furthermore, bubble growth suppresses with high surface tension. Moreover, bubbles coalesce and merging may not take place upon collision since a strong cohesive force between interacting bubbles, which may avoid the formation of the slug flow due to the merging of bubbles.

Liquid viscosity also plays an important role in the process of boiling in the microchannel. The higher liquid viscosity gives rise to the drag or shear effect experienced by the bubble anchored at the nucleation site. Thus, bubble distortion at the nucleation site is likely to be governed by viscosity, leading to the early departure of the bubble. Furthermore, the resistance between the wall-liquid and liquid-elongated bubble increases with an increase in the viscosity. This reduces the velocity of the elongated bubble and increases the residence time of the bubble inside the microchannel. If latent heat of vaporization is low and heat flux is sufficiently high, rapid evaporation takes place for a longer time (longer resident time at high viscosity) that may lead to dry out of the film. Otherwise, elongated bubble would escape easily from the microchannel outlet with high velocity.

Low vapor density is responsible for the bubble lift off from the nucleation site. On the other hand, liquid column above the bubble at the nucleation site opposes the lifting off the bubble in case of high liquid density. Thus, fluid with low liquid density and high vapor density is desirable for the easy detachment of the bubble from the nucleation site. Due to low vapor density, bubble might lift off in the bulk liquid stream after detachment without sliding on the heat transfer surface (microchannel wall). However, bubble sliding after detachment in microchannel has not yet been confirmed.

8 Conclusion

A detailed literature review relevant to bubble dynamics in micro-channel has been carried out and here are the concluding remarks/future suggestions.

1. Bubble dynamics in the microchannel have been extensively studied experimentally and confirmed the existence of the thin liquid film, which is trapped between the vapor bubble and the microchannel wall. On the other hand, few researchers reported that bubble nucleate grow and detach without experiencing confinement. Thus, it is still unclear to identify dominating heat transfer mechanism, nucleate boiling or evaporation of thin liquid film. This is even more complicated to identify at vigorous boiling (at low mass flux or high heat flux) and led the topic of intense research.
2. Prediction of the bubble growth rate nucleated bubble or elongated bubble in the microchannel is important from the thermal and hydraulic analysis point of view. The previously proposed models show the linear variation of the bubble growth at the nucleation site and hence further attention need to focus to proposed the models which would be capable to account for bubble growth regimes at the nucleation site. This can be achieved by considering the thermal boundary layer and its behavior near the wall the during bubble ebullition cycle. Further, it is equally important to check the validity of the previous model for other fluids (such as refrigerants, HFE and FC series fluids) flowing through a microchannel. Furthermore, modelling of confined and elongated vapor bubble has been carried out but it is observed that the presence and prediction of initial liquid film thickness in the microchannel heat sink needs special attention.
3. Bubble in microchannels grows under the influence of various effects such as surface tension,

inertia, shear, gravitation/buoyancy and evaporation momentum. The dominance of these forces during different stage of bubble growth (such as nucleated bubble, confined bubble, bubble departure) need to be studied extensively. Furthermore, forces induced due to wall confinement plays a great role hence its quantitative analysis need to be carryout. Among these effects, some try to hold the bubble at the nucleation site and some try to detach the bubble. It can be possible to obtained bubble departure diameter by carefully balancing (holding and detaching) the resistive effects and possible additional effects.

4. The impact of the thermophysical properties on the bubble dynamics associated with microchannel has qualitatively been discussed. However, the combined effect of these thermophysical properties on the flow boiling in the microchannel are expectedly complicated and needs further attention. Open literature indicates that unlike water, bubble dynamics using refrigerant or other fluids have not been extensively studied. Such a study can be helpful to provide more insight into the flow boiling phenomena in the microchannel.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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